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Profile

Engineering student at **CentraleSupélec / Université Paris-Saclay**, focused on **ML research, optimization, and data-driven systems**. Experienced in building end-to-end ML pipelines, with strong foundations in **statistics, convex optimization, NLP, and computer vision**.

Education

CentraleSupélec, Gif-sur-Yvette

Sep 2023 – Present

- **Mathematics:** Probability (CIP), Advanced Statistics, Optimization, PDEs.
- **CS & Data:** Algorithms & Complexity, Big Data, Web Data Intelligence.
- **Projects:** Modeling (Biotech & Health), Structural optimization for additive manufacturing, Blockchain.

Lycée Louis-le-Grand – Classes Préparatoires MP2I/MPI*

Sep 2021 – Jul 2023

Experience & Projects

AppliedAI

Mar – Jun 2026

Machine Learning Engineer

- Developed **tools** for a **multimodal planning agent** within the Opus product.
- Designed an **agent playground**: a **learning** pipeline embedded in agent workflows, with an **LLM-in-the-loop** for assisted creation and iteration of new agents.
- Built a **few-shot learning** system: embeddings of structured multimodal inputs (JSON) and **semantic search** over human-validated responses on **Weaviate**.
- **Automated test generation** from human reviews, producing evaluation suites for continuous agent validation.
- Implemented the **GEPA** (Genetic-Pareto) algorithm: **reflective prompt optimization** and agent evolution driven by execution feedback.

Paris Digital Lab – CentraleSupélec

Sep 2025 – Feb 2026

Selected among 23 students for a 6-month program on digital innovation.

Optimization – TotalEnergies

Sep – Nov 2025

- Formulated a **convex optimization** problem and solved it with a non-linear solver (Hexaly) under polynomial constraints.
- Benchmarked **RRT***, **Dijkstra**, **A***; designed a state-of-the-art algorithm based on a **Visibility Road Map (VRM)** with path smoothing.

Next Best Product Prediction – Palatine (BPCE)

Nov – Dec 2025

- **Time-series analysis** over several years of large-scale banking client history.
- **NLP & GenAI:** leveraged LLM embeddings to isolate weak signals in heterogeneous data.
- Built a **hybrid ML pipeline (LightGBM)** combining classification and regression; rigorous evaluation (Precision-Recall).
- **Explainability:** applied **SHAP** to ensure transparent recommendations.

Flood Detection & Prevention – Matmut

Dec 2025 – Feb 2026

- Implemented a **YOLO26** model for flood detection; built an annotated dataset (**20+ classes**, Roboflow).
- Tested ML Sharp models for augmented-reality simulation; prototyped **real-time** visualization of rising water levels.

73Strings – AI Engineer Intern

Sep – Dec 2025

- Designed a **GraphRAG** system from scratch: entity extraction and hierarchical document organization.
- Implemented it with **Neo4j**, **Cypher**, and **LangChain** for LLM orchestration; OCR for information extraction.

CNRS – ENSIIE & MICS

Jan – Jun 2025

Research project – Gaussian graphical models in genetics

- Developed an inference method for the **Graphical Lasso** via **convex optimization (ADMM)**.
- Implemented the SRMDG and JEGN variants; applied to **50 GB of TCGA genetic data** using **Kendall's Tau** correlation.

Mistral AI MCP Hackathon (1st place) – Paris

Jul 2025

- Built an **MCP** server integrating the Strava API, performance analytics, and training-plan generation.
- Deployed an **end-to-end** system with Ngrok and Hugging Face; ranked **#1** out of 35+ teams.

Technical Skills

Languages: Python, C++, C, OCaml, MATLAB, JavaScript

ML/DL: scikit-learn, PyTorch, LightGBM, SHAP, YOLO26, LangChain

Data/Infra: SQL, Neo4j, Weaviate, Git, Jupyter, REST APIs, Hexaly

Tools: React, Kubernetes, Roboflow, Lens Studio

Languages

French (native), English (fluent), Spanish (intermediate), Mandarin (basic)

Interests

Piano (ABRSM Grade 7), tennis (French ranking 15/3), mountaineering, skiing, windsurfing